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Answer: 1

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & k & 1 \end{bmatrix}$$

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$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ k & 0 & 1 \end{bmatrix}$$

N 33

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, E = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

$$(\det E)(\det A) = (1-0)(a d - b c) = a d - b c$$

$$\det EA = \begin{bmatrix} a+kc & b+kd \\ c & d \end{bmatrix} =$$

$$= (a+kc)d - (b+kd)c = ad + kcd - bc - kcd = ad - bc$$

$$\text{Answer: } \det EA = (\det E)(\det A)$$

N 35

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$(\det E)(\det A) = (0-1)(ad-bc) = bc - ad$$

$$\det EA = \begin{bmatrix} c & d \\ a & b \end{bmatrix} = bc - ad$$

$$\text{Answer: } \det EA = (\det E)(\det A)$$

N 37

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} \text{ and } 5A = \begin{bmatrix} 15 & 5 \\ 20 & 10 \end{bmatrix}$$

$$\det 5A = 50$$

$$5 \det A = 10$$

$$\text{Answer: } \det 5A = 5 \det A$$

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$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = 4 \begin{bmatrix} a & b & c \\ d & e & f \\ 3g & 3h & 3i \end{bmatrix} = 24 \quad \text{Answer: 24}$$

Q24

$$\begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix} \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix} \begin{bmatrix} -3 \\ 5 \\ -2 \end{bmatrix} \quad \det \begin{bmatrix} 4 & -1 & -3 \\ 6 & 0 & -5 \\ 2 & 7 & -2 \end{bmatrix} = 0$$

$\det = 0 \Rightarrow$ invertible $\Rightarrow$ dep

Ans: vectors are linearly dependent

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Q3

$$\det A = \begin{vmatrix} 1 & 1 & 0 \\ -3 & 0 & 2 \\ 0 & 1 & -2 \end{vmatrix} = -8$$

$$\det A_1(b) = \begin{vmatrix} 3 & 1 & 0 \\ 0 & 0 & 2 \\ 2 & 1 & -2 \end{vmatrix} = -2$$

$$\det A_2(b) = \begin{vmatrix} 1 & 3 & 0 \\ -3 & 0 & 2 \\ 0 & 2 & -2 \end{vmatrix} = -22$$

$$\det A_3(b) = \begin{vmatrix} 1 & 1 & 3 \\ -3 & 0 & 0 \\ 0 & 1 & 2 \end{vmatrix} = -3$$

$$x_1 = \frac{\det(A_1(b))}{\det A} = \frac{1}{4}$$

$$x_2 = \frac{\det(A_2(b))}{\det A} = \frac{-11}{4}$$

$$x_3 = \frac{\det(A_3(b))}{\det A} = \frac{3}{8}$$

Answer: $x_1 = \frac{1}{4}, x_2 = \frac{-11}{4}, x_3 = \frac{3}{8}$

Q4

$$65x_1 + 4x_2 = 5$$

$$3x_1 + 25x_2 = -2$$

$$\begin{bmatrix} 65 & 4 \\ 3 & 25 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}$$

$$\det \begin{bmatrix} 65 & 4 \\ 3 & 25 \end{bmatrix} = 125^2 - 36$$

$$S = \pm \sqrt{3}$$

$$\det A(b) = \begin{vmatrix} 5 & 4 \\ -2 & 2s \end{vmatrix} = 10s + 8$$

$$\det A_2(b) = \begin{vmatrix} 6s & 5 \\ 3 & -2 \end{vmatrix} = 12s - 4s$$

Answers

$$x_1 = \frac{\det A_1(b)}{\det A} = \frac{5s+4}{6(s^2-3)}$$

$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{-4s-15}{4(s^2-3)}$$

Ans

$$1x_1 + 2sx_2 = -1$$

$$3x_1 + 6sx_2 = 4$$

$$\begin{bmatrix} 1 & 2s \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$\det \begin{bmatrix} 1 & 2s \\ 3 & 6 \end{bmatrix} = 6s^2 - 6s \neq 0$$

$$6s(s-1) \neq 0$$

$$s \neq 0, s \neq 1$$

invertible

$$\det A_1(b) = \begin{vmatrix} -1 & 2s \\ 4 & 6s \end{vmatrix} = -6s - 8s$$

$$\det A_2(b) = \begin{vmatrix} 5 & -1 \\ 3 & 4 \end{vmatrix} = 4s + 3$$

A is invertible

$$x_1 = \frac{\det A_1(b)}{\det A} = \frac{-6s-8s}{6s(s-1)} = \frac{-14s}{3s(s-1)}$$

$$x_2 = \frac{\det A_2(b)}{\det A} = \frac{4s+3}{6s(s-1)}$$

Answer: $x_1 = \frac{-14s}{3s(s-1)}, x_2 = \frac{4s+3}{6s(s-1)}$